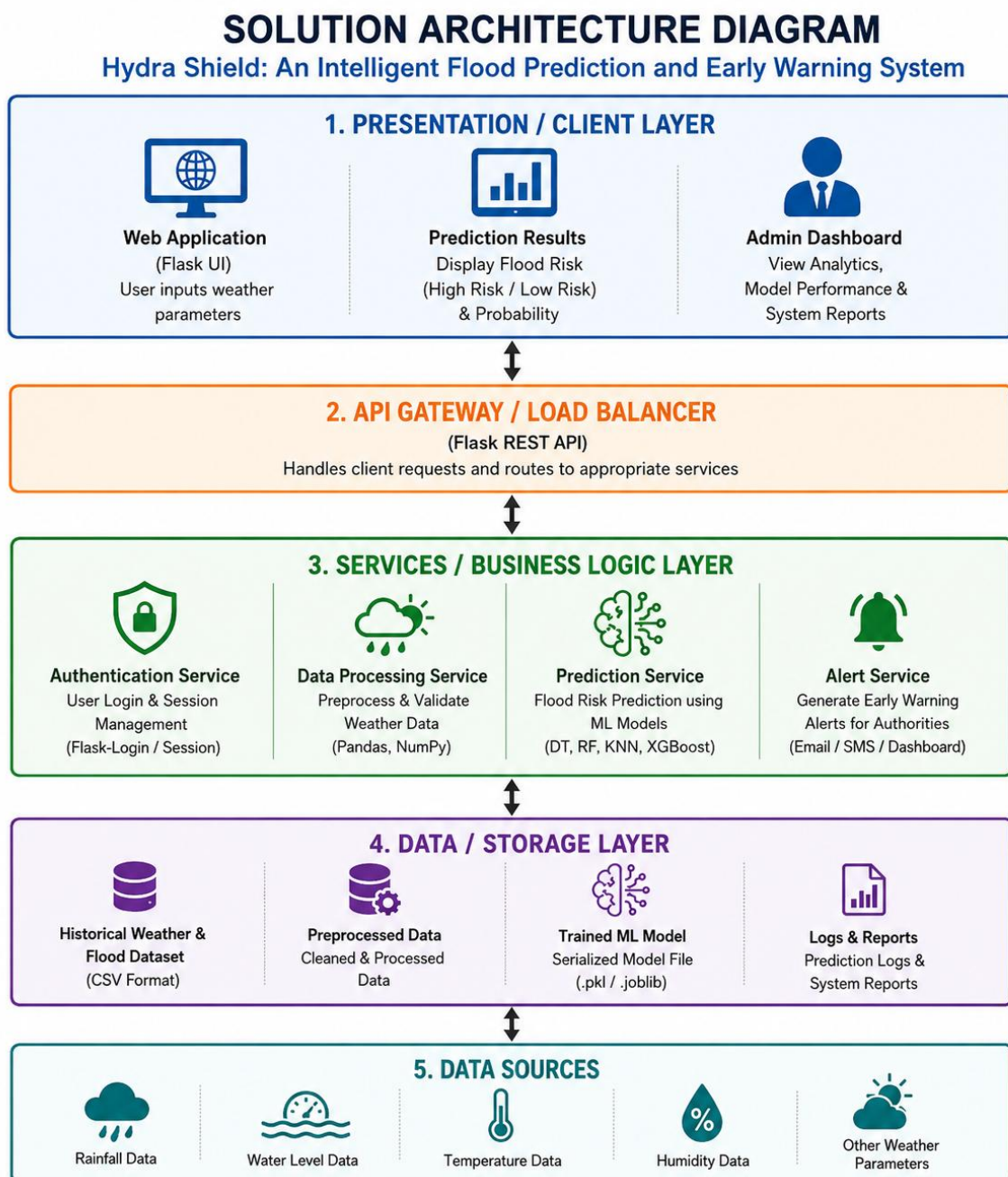


Solution Architecture

Date	28 June, 2026
Team ID	
Project Name	Hydra Shield: An Intelligent Flood Prediction and Early Warning System (Rising Waters)
Maximum Marks	5 marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description/Role	Technologies Used
Presentation Layer	Takes weather details from the user and shows the flood prediction result.	Flask, HTML, CSS
API Gateway	Sends user requests to the prediction system.	Flask
Prediction Service	Processes the data and predicts the flood risk.	Python, Scikit-learn, XGBoost
Data Storage	Stores the flood dataset and trained model.	CSV, Pickle